

Team 4 Project: HR Employee Attrition Prediction Using Machine Learning

Team Members

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Abstract

Employee attrition is a major challenge for organizations because high turnover increases recruitment costs, training expenses, and productivity loss. This project focused on predicting employee attrition using machine learning techniques on the IBM HR Employee Attrition dataset. The dataset contains demographic, work-related, and satisfaction-related information for 1,470 employees. Exploratory Data Analysis (EDA), preprocessing, oversampling, and predictive modeling were performed using Logistic Regression, Decision Tree, and Random Forest models. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and Area Under the Curve (AUC). Logistic Regression achieved the best overall business performance with a Recall of 81.69% and an AUC of 0.8336, making it the most suitable model for identifying employees likely to leave the organization. The findings of this project can help HR departments make data-driven decisions to improve employee retention strategies.

1. Introduction

Employee attrition refers to employees leaving an organization through resignation, retirement, or termination. High attrition negatively affects organizational productivity, increases recruitment costs, and creates operational instability. Organizations therefore aim to identify employees at risk of leaving before attrition occurs.

Machine learning techniques can help organizations analyze employee data and predict attrition patterns. Predictive analytics allows HR departments to identify factors contributing to employee turnover and develop targeted retention strategies.

The objective of this project is to develop predictive models capable of identifying employees at risk of attrition using the IBM HR Employee Attrition dataset. The project also compares multiple machine learning algorithms to determine the most effective model for attrition prediction.

2. Methods

2.1 Dataset Description

The IBM HR Employee Attrition dataset was used for this project. The dataset contains employee demographic information, work-related variables, compensation details, and employee satisfaction metrics.

Dataset Information	Value
Number of Rows	1470
Number of Columns	35
Target Variable	Attrition

The target variable contains two classes:

- Yes – Employee left the organization
- No – Employee remained in the organization

2.2 Data Cleaning and Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning analysis. Columns such as EmployeeNumber, EmployeeCount, Over18, and StandardHours were removed because they did not provide meaningful predictive information.

Categorical variables were converted into factor variables, and one-hot encoding was applied using the `model.matrix()` function to transform categorical variables into numerical format suitable for machine learning algorithms.

The target variable Attrition was converted into binary form:

- 0 = No Attrition
- 1 = Attrition

The dataset was then divided into training and testing datasets using a 70:30 split.

2.3 Handling Class Imbalance

The dataset contained a significantly larger number of employees without attrition compared to employees with attrition. This imbalance could cause machine learning models to favor the majority class.

To address this issue, oversampling was applied to the minority class in the training dataset. Minority class observations were duplicated until both classes contained equal numbers of observations. This improved the model's ability to identify employees likely to leave the organization.

2.4 Exploratory Data Analysis (EDA)(Appendix A)

Exploratory Data Analysis was conducted to identify important trends and relationships associated with employee attrition.

Attrition Distribution

The dataset showed class imbalance, with approximately 84% of employees remaining in the organization and only 16% leaving.

Attrition by Department(Appendix A- (i))

The Sales and Human Resources departments experienced higher attrition rates compared to Research & Development.

Age and Attrition(Appendix A- (ii))

Younger employees demonstrated higher attrition rates compared to older employees.

Monthly Income and Attrition(Appendix A- (iii))

Employees with lower monthly income showed a higher probability of attrition.

Job Satisfaction and Attrition(Appendix A- (iv))

Employees with lower job satisfaction levels had significantly higher attrition rates.

Overtime and Attrition(Appendix A- (v))

Employees working overtime experienced substantially higher attrition rates than employees who did not work overtime.

Correlation Analysis(Appendix A- (vi))

The correlation heatmap indicated strong relationships among variables such as MonthlyIncome, JobLevel, and TotalWorkingYears. These relationships suggested potential multicollinearity among predictors.

2.5 Machine Learning Models

Three machine learning models were developed and evaluated:

Logistic Regression(Appendix B- (i))

Logistic Regression was selected because of its interpretability and effectiveness for binary classification problems.

Decision Tree(Appendix B- (ii))

Decision Tree classification was used to identify hierarchical decision rules contributing to employee attrition.

Random Forest(Appendix B- (iii))

Random Forest was implemented as an ensemble learning approach using multiple decision trees.

2.6 Threshold Tuning(Appendix C- (i))

Threshold tuning was performed on the Logistic Regression model by testing probability thresholds ranging from 0.10 to 0.90.

Since the primary business objective was to identify employees likely to leave, Recall was prioritized over Accuracy. A threshold of 0.50 was selected because it maintained strong Recall performance while also producing the highest F1-Score among high-recall thresholds.

2.7 Cross Validation(Appendix C- (ii))

To evaluate model robustness and reduce overfitting, 10-fold cross validation was applied to the Logistic Regression model. The training dataset was divided into ten folds, and the model was repeatedly trained and validated across multiple iterations.

The cross-validation results demonstrated stable model performance, indicating that the model generalized effectively to unseen data.

3. Results

3.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	76.82%	39.46%	81.69%	53.21%	0.8336
Decision Tree	72.73%	31.58%	59.15%	41.18%	0.6890
Random Forest	85.23%	63.64%	19.72%	30.11%	0.8340

The Logistic Regression model achieved the highest Recall among all models, making it the most effective model for identifying employees likely to leave the organization.

Although Random Forest achieved the highest Accuracy, its Recall was very low, meaning that many actual attrition cases were not identified. Since attrition prediction prioritizes identifying employees likely to leave, Recall was considered more important than Accuracy alone.

Decision Tree produced weaker overall performance compared to Logistic Regression and Random Forest.

3.2 ROC Curve Analysis(Appendix D)

ROC curve analysis was conducted to compare the classification performance of all three models.

Logistic Regression and Random Forest achieved similar AUC values of approximately 0.83, indicating strong classification capability. However, Logistic Regression achieved significantly better Recall performance, making it more suitable for the business objective of employee attrition prediction.

Decision Tree achieved a lower AUC value of 0.6890, indicating weaker predictive capability.

3.3 Business Insights

Several important business insights were identified during the analysis:

- Employees working overtime showed significantly higher attrition rates.
- Lower monthly income was associated with increased employee turnover.
- Employees with lower job satisfaction were more likely to leave the organization.
- Younger employees experienced higher attrition rates.
- Work-life balance and environment satisfaction influenced employee retention.

The Decision Tree model further indicated that overtime, monthly income, years with current manager, and stock option level were major factors influencing attrition.

4. Conclusion

This project successfully developed machine learning models for predicting employee attrition using the IBM HR Employee Attrition dataset. Logistic Regression, Decision Tree, and Random Forest models were evaluated using multiple performance metrics.

Logistic Regression was selected as the final model because it achieved the highest Recall while maintaining strong overall classification performance. Since employee attrition prediction focuses on identifying employees likely to leave, Recall was considered the most important evaluation metric.

The analysis demonstrated that overtime, monthly income, job satisfaction, and work-life balance strongly influence employee attrition. Organizations can use these insights to implement targeted retention strategies, improve employee satisfaction, and reduce workforce turnover.

Future improvements may include advanced oversampling techniques such as SMOTE, hyperparameter tuning, and ensemble learning approaches to further improve model performance.

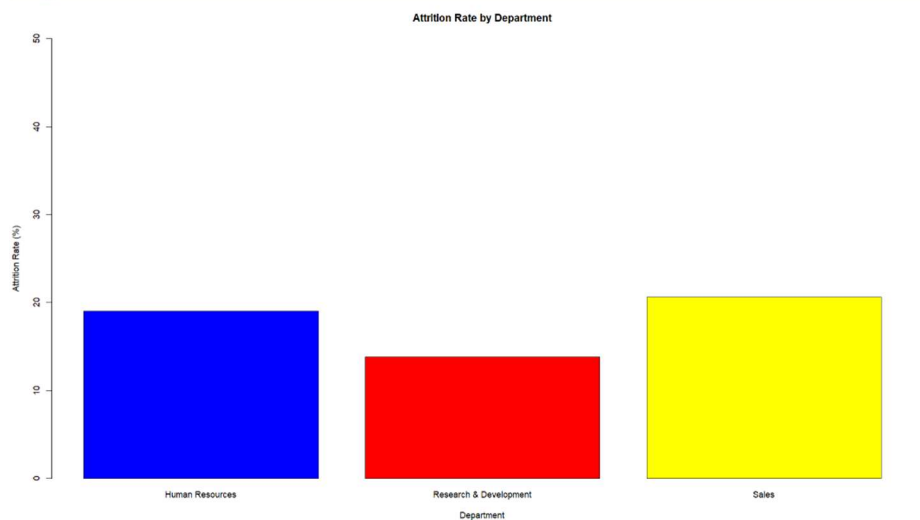
5. References

1. IBM HR Analytics Employee Attrition Dataset
2. Kuhn, M. Caret Package Documentation
3. Breiman, L. Random Forests, Machine Learning Journal
4. R Documentation for Logistic Regression, Decision Trees, and Random Forest Algorithms

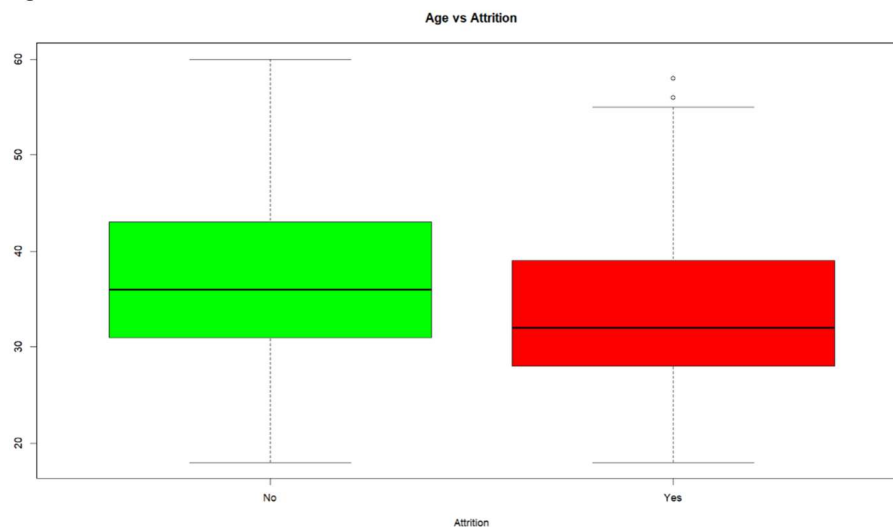
6. Appendices

Appendix A – Additional Visualizations

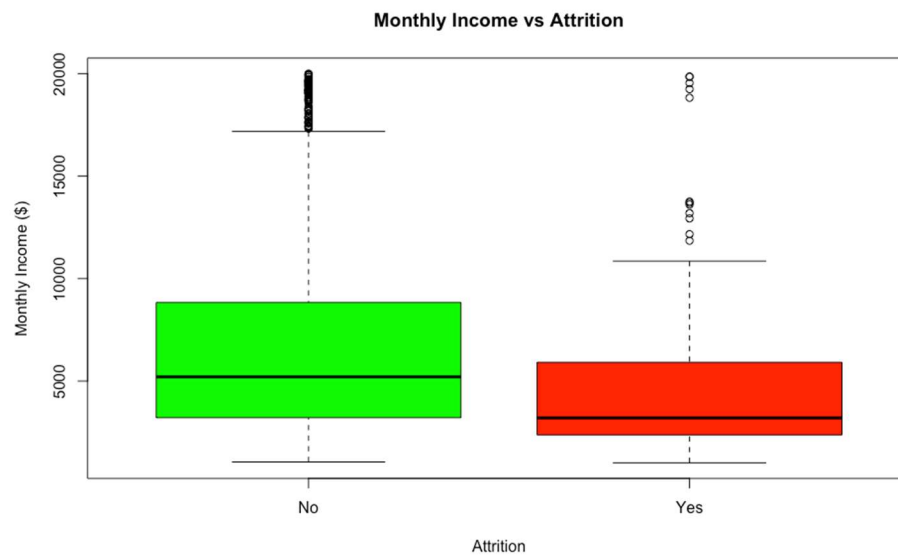
(i) Department Vs Attrition



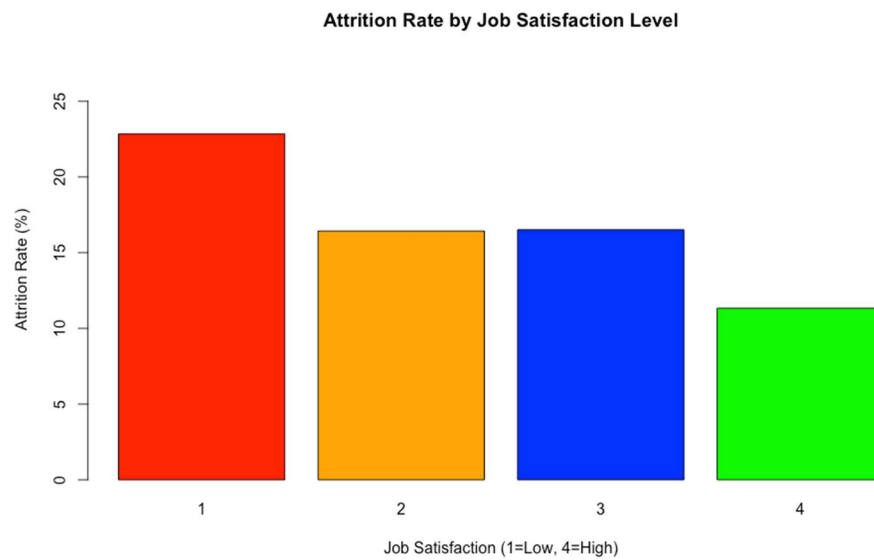
(ii) Age Vs Attrition



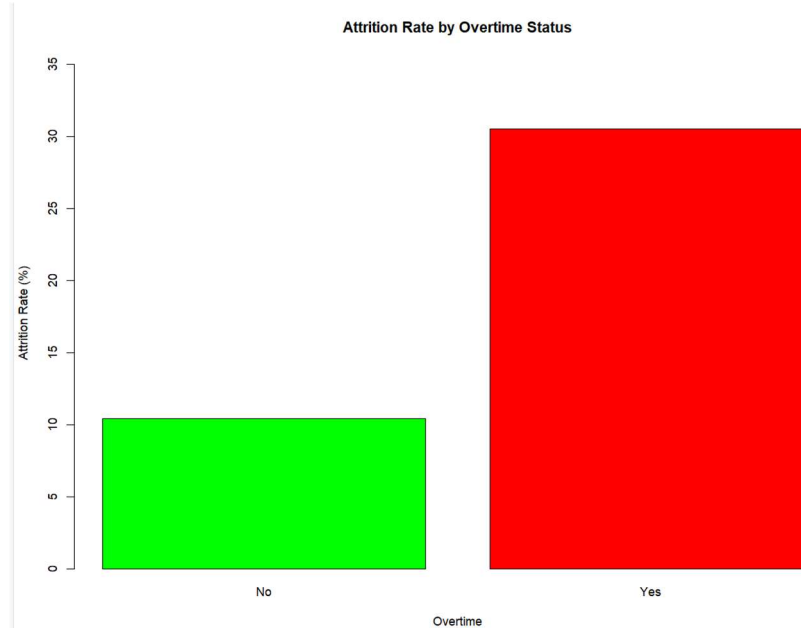
(iii) Income Vs Attrition



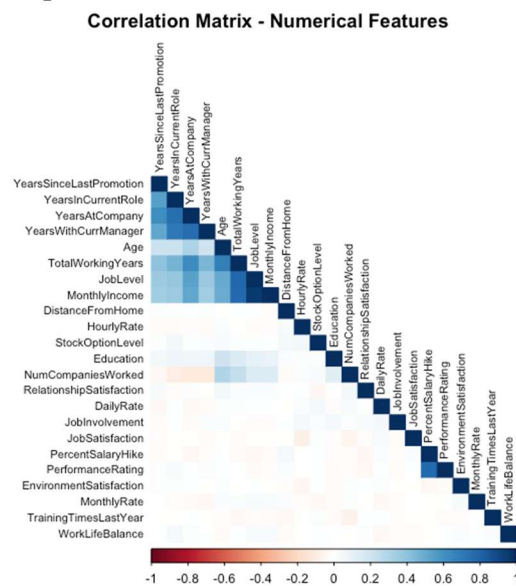
(iv) Job Satisfaction Vs Attrition



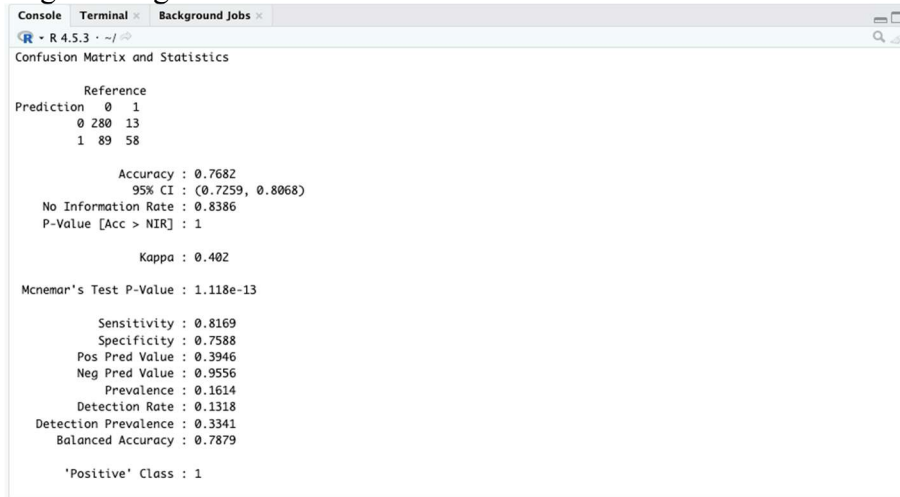
(v) Overtime Vs Attrition



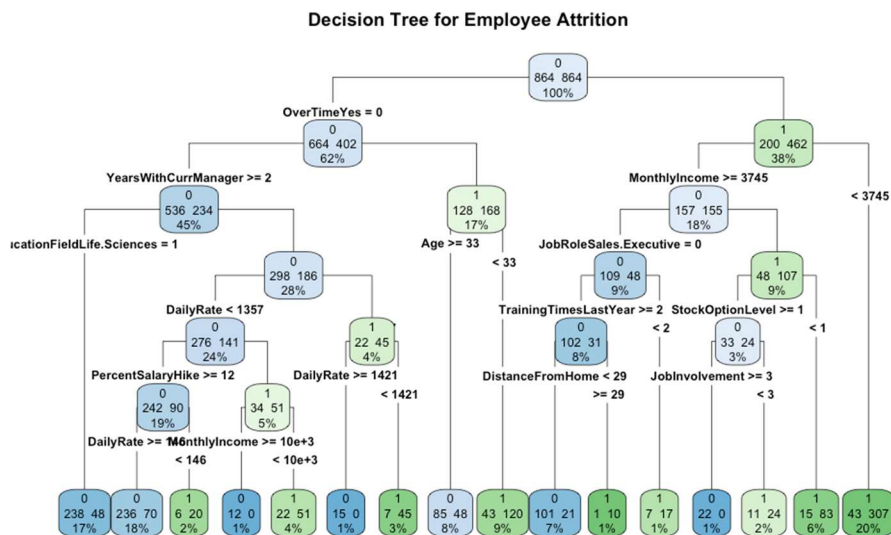
(vi) Correlation Heatmap



(i) Logistic Regression Confusion Matrix



(ii) Decision Tree
→ Decision Tree Plot



→ Decision Tree Confusion Matrix

```
Console | Terminal | Background Jobs |
R • R 4.5.3 • ~/
Confusion Matrix and Statistics

      Reference
Prediction 0  1
0    278  29
1     91  42

      Accuracy : 0.7273
      95% CI   : (0.6831, 0.7684)
No Information Rate : 0.8386
P-Value [Acc > NIR] : 1

      Kappa : 0.255

McNemar's Test P-Value : 2.569e-08

      Sensitivity : 0.59155
      Specificity : 0.75339
      Pos Pred Value : 0.31579
      Neg Pred Value : 0.90554
      Prevalence : 0.16136
      Detection Rate : 0.09545
      Detection Prevalence : 0.30227
      Balanced Accuracy : 0.67247

'Positive' Class : 1
```

(iii) Random Forest

→ Random Forest Confusion Matrix

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Confusion Matrix and Statistics

      Reference
Prediction 0  1
0    361  57
1      8  14

      Accuracy : 0.8523
      95% CI   : (0.8156, 0.8841)
No Information Rate : 0.8386
P-Value [Acc > NIR] : 0.2402

      Kappa : 0.2433

McNemar's Test P-Value : 2.622e-09

      Sensitivity : 0.19718
      Specificity : 0.97832
      Pos Pred Value : 0.63636
      Neg Pred Value : 0.86364
      Prevalence : 0.16136
      Detection Rate : 0.03182
      Detection Prevalence : 0.05000
      Balanced Accuracy : 0.58775

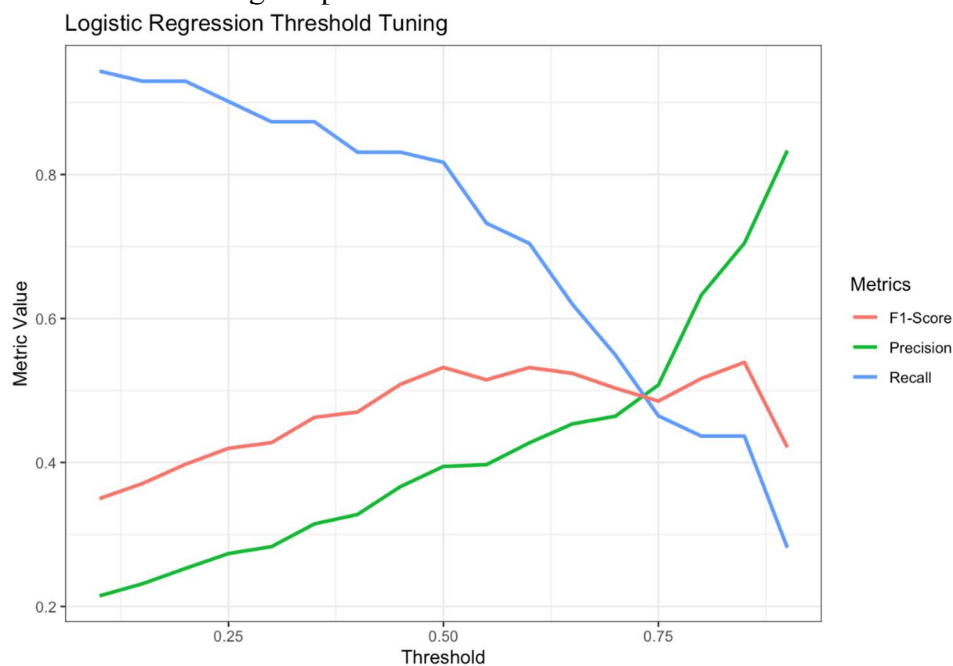
'Positive' Class : 1
```

Appendix C

(i) Threshold Tuning → Full Threshold Table

	Threshold	Accuracy	Precision	Recall	Specificity	F1_Score
Accuracy	0.10	0.4340909	0.2147436	0.9436620	0.3360434	0.3498695
Accuracy1	0.15	0.4909091	0.2315789	0.9295775	0.4065041	0.3707865
Accuracy2	0.20	0.5454545	0.2528736	0.9295775	0.4715447	0.3975904
Accuracy3	0.25	0.5977273	0.2735043	0.9014085	0.5392954	0.4196721
Accuracy4	0.30	0.6227273	0.2831050	0.8732394	0.5745257	0.4275862
Accuracy5	0.35	0.6727273	0.3147208	0.8732394	0.6341463	0.4626866
Accuracy6	0.40	0.6977273	0.3277778	0.8309859	0.6720867	0.4701195
Accuracy7	0.45	0.7409091	0.3664596	0.8309859	0.7235772	0.5086207
Accuracy8	0.50	0.7681818	0.3945578	0.8169014	0.7588076	0.5321101
Accuracy9	0.55	0.7772727	0.3969466	0.7323944	0.7859079	0.5148515
Accuracy10	0.60	0.8000000	0.4273504	0.7042254	0.8184282	0.5319149
Accuracy11	0.65	0.8181818	0.4536082	0.6197183	0.8563686	0.5238095
Accuracy12	0.70	0.8250000	0.4642857	0.5492958	0.8780488	0.5032258
Accuracy13	0.75	0.8409091	0.5076923	0.4647887	0.9132791	0.4852941
Accuracy14	0.80	0.8681818	0.6326531	0.4366197	0.9512195	0.5166667
Accuracy15	0.85	0.8795455	0.7045455	0.4366197	0.9647696	0.5391304
Accuracy16	0.90	0.8750000	0.8333333	0.2816901	0.9891599	0.4210526

→ Threshold Tuning Graph



(ii) Cross Validation

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Generalized Linear Model

1728 samples
44 predictor
2 classes: '0', '1'

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 1554, 1554, 1556, 1556, 1556, 1555, ...
Resampling results:

Accuracy Kappa
0.774345 0.5487466
```

Appendix D

ROC Curve Comparison

